

Capstone Project

Alex Paquette

February 4th 2026



OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



Summarize key findings here

- **Web and Cloud technologies** have the most used and desired
 - Top languages include the following (JavaScript, HTML/CSS, TypeScript, and PHP)
 - Top Platforms list the following (AWS, Azure, Google Cloud)
 - These are all web-based technologies
- There is **consistency between** what technology is used and is desired
 - There is little variation between technologies being used, and technologies desired
 - This suggests there will likely be little change in what technologies are used in the near future
- The vast majority of respondents are **above 25 years old**
 - This suggests an aging developer demographic, which aligns with evidence that there are less entry-level positions



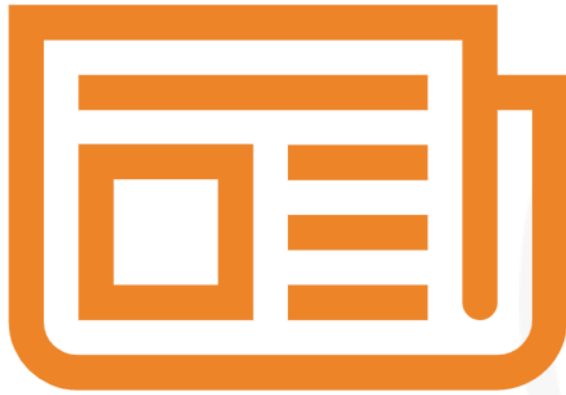
INTRODUCTION



Purpose of the report, Target Audience & Value of the analysis

- This report is aimed at Data Analyst
- It is meant to inform professionals on current and future technology trends
- These insights can inform developers and analysts on what technologies they should learn for future job security

METHODOLOGY

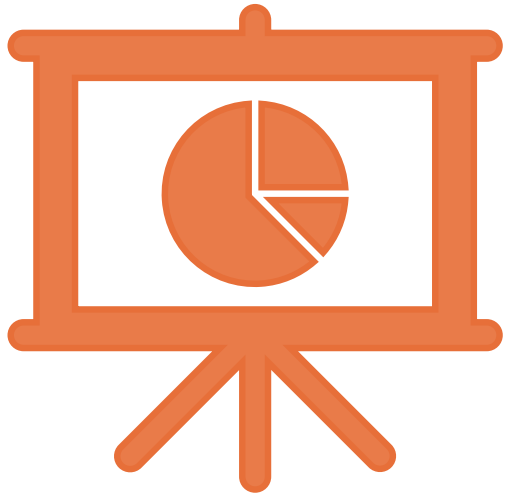


Data sources, Data collection & Data wrangling steps

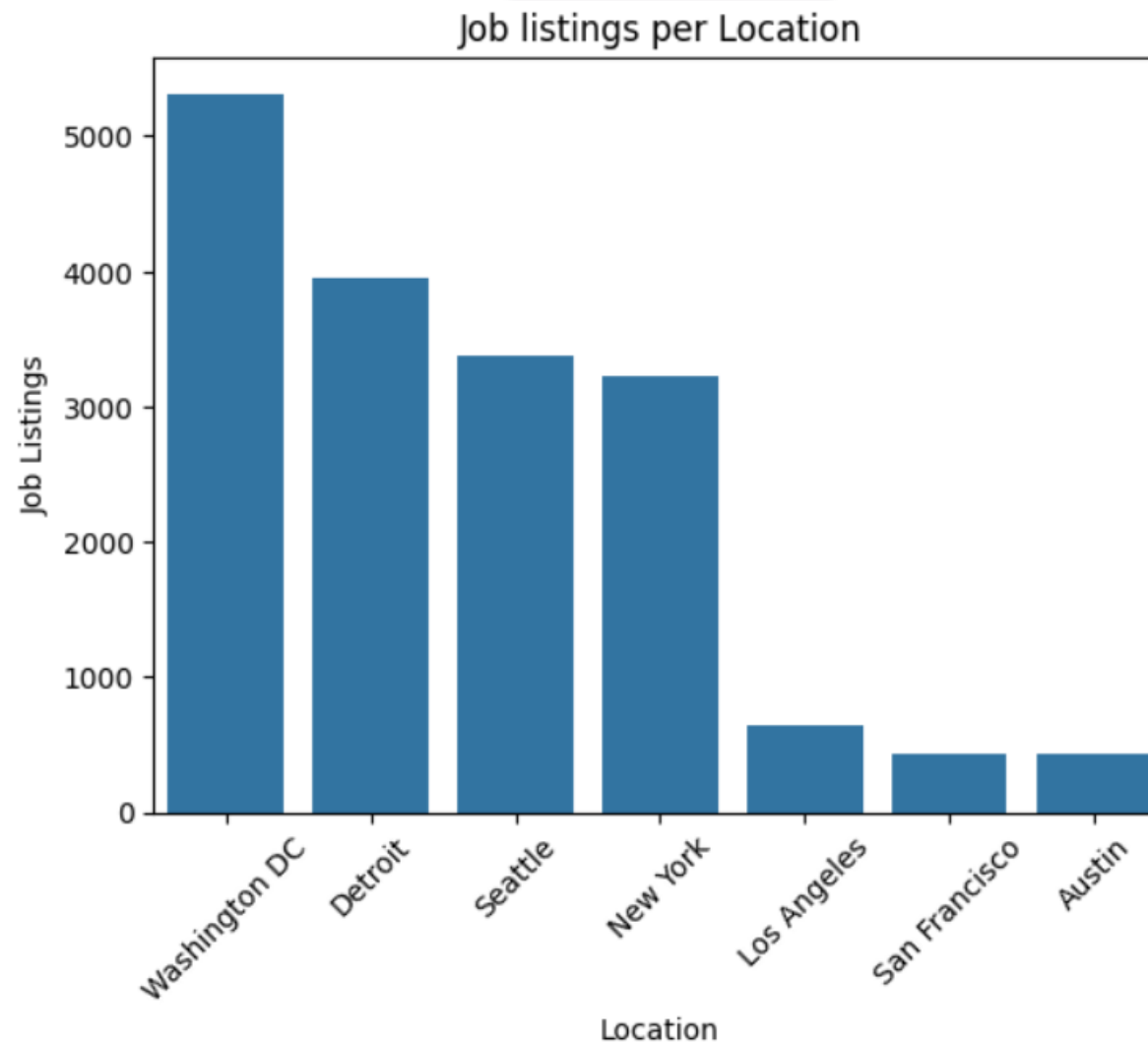
- Data sources
 - Identify relevant data domains
 - Access sources reliably
 - Define data access
- Data Collection
 - Extract raw data
 - Validate extraction
 - Store raw data
- Data Wrangling
 - Data cleaning (nulls, normalization, etc)
 - Data transformation (derive new insights from data)
- Data Exploration
 - Produce visualizations
 - Draw conclusions



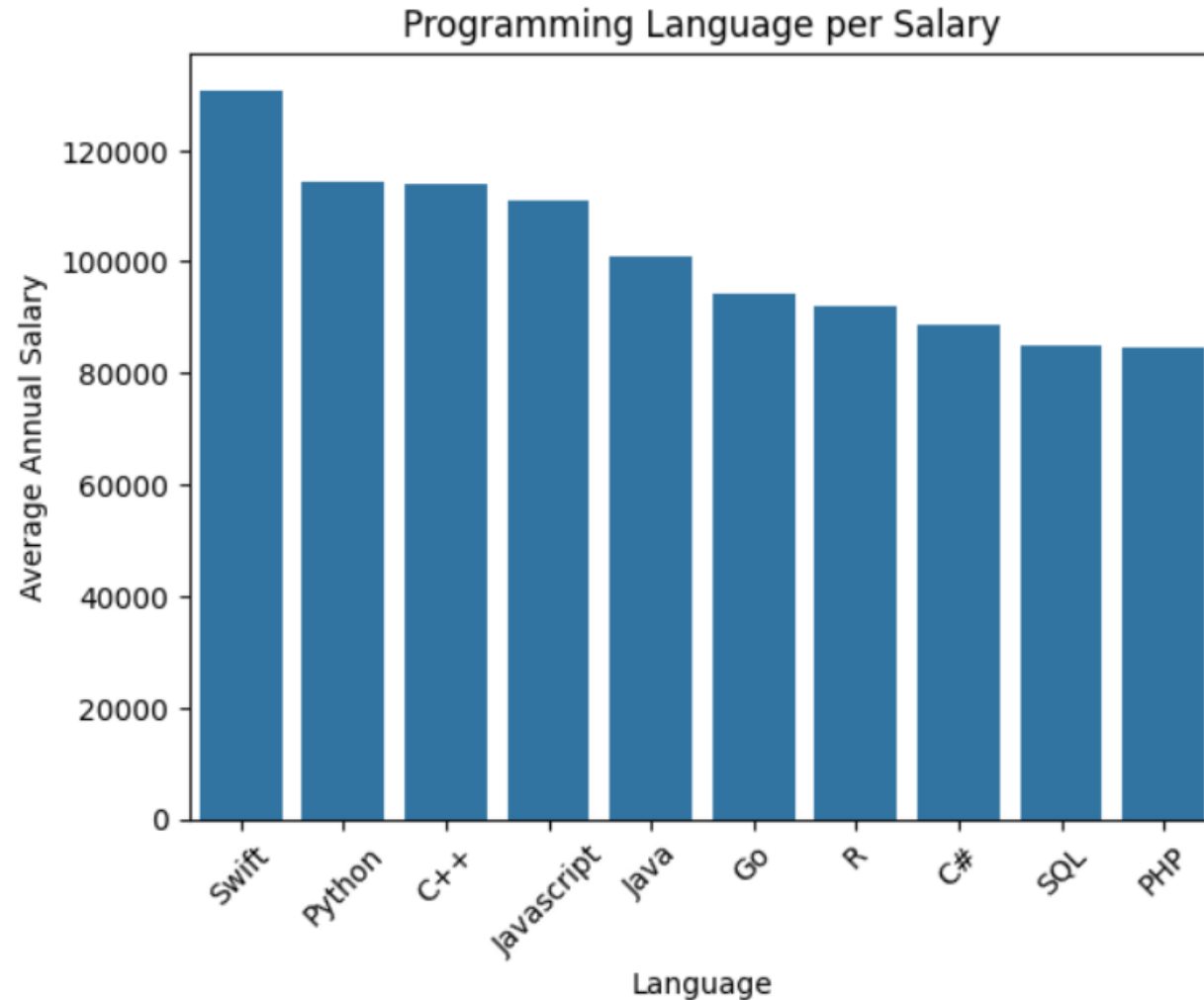
Results



JOB POSTINGS



POPULAR LANGUAGES

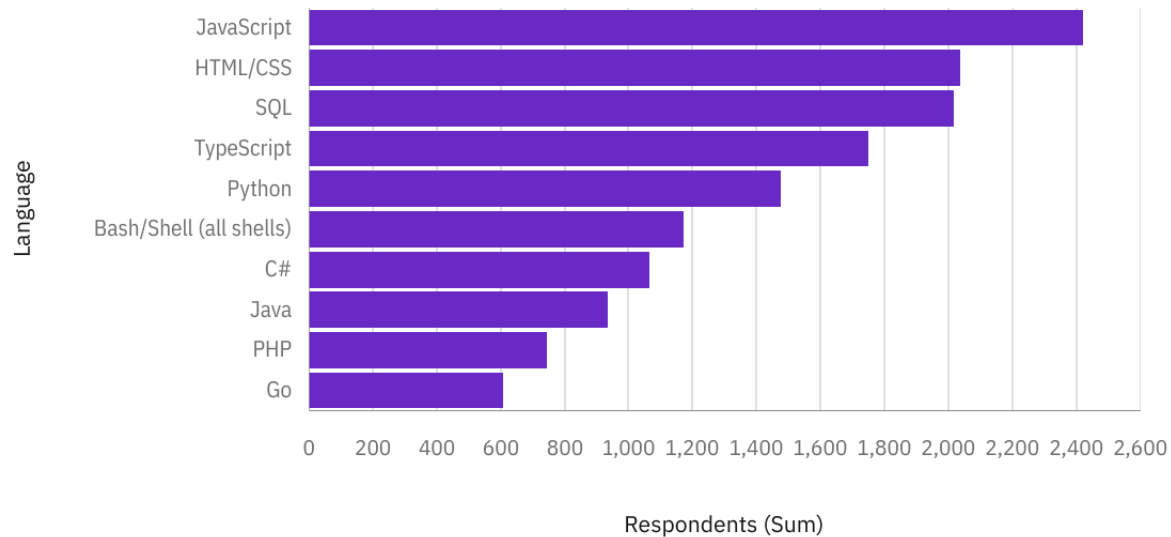


PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

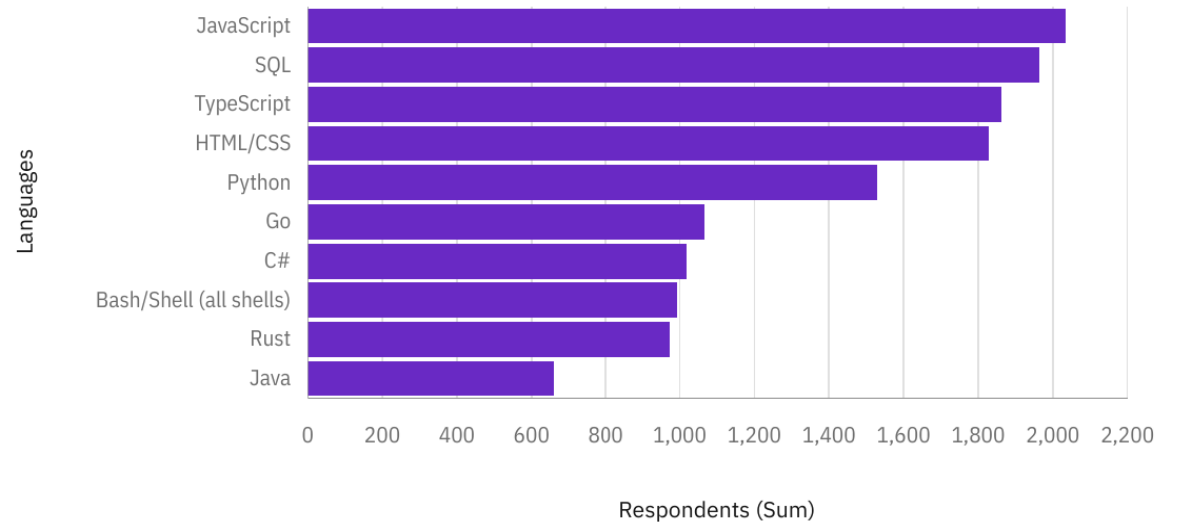
Current Year

Top 10 Programming Languages Used



Next Year

Top 10 Desired Programming Languages



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- The majority of the languages are related to **web development**
- Most languages remain the same **in both charts**
- **Rust** makes an appearance in the desired chart, while not being widely used

Implications

- The wide use and desirability of web-based technologies suggest a continued shift towards **cloud solution**
- Rust's appearance on this chart suggest the language is likely to **increase in popularity**

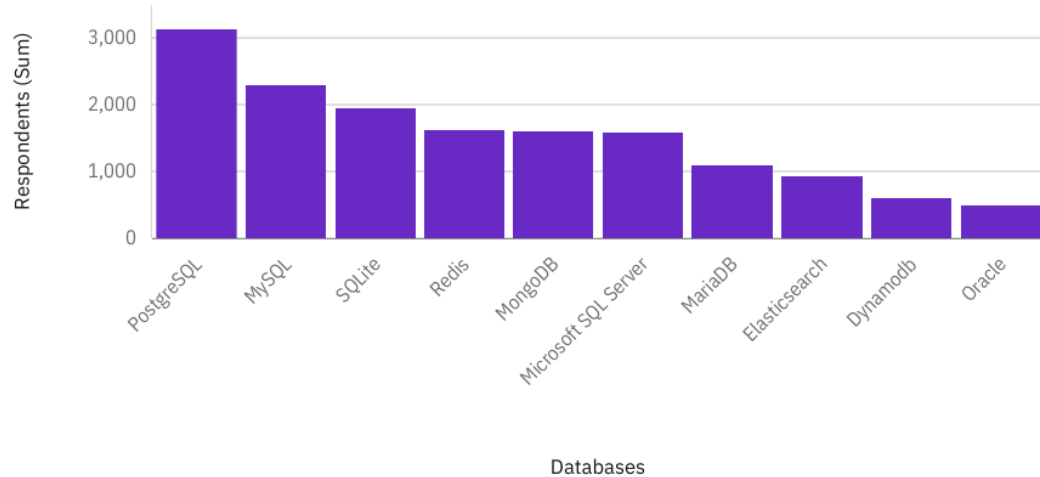


DATABASE TRENDS

Summarize key trends shown in the charts.

Current Year

Top 10 Databases Used



Next Year

Top 10 Desired Databases



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- **PostgreSQL** is both highly used and highly desired
- Most databases listed remain the **same in both charts**
- **Supabase** makes an appearance in the desired chart, while not being widely used

Implications

- The current trend suggests **stable database technologies**
- PostgreSQL being highly used, and even more highly desired suggests **its usage may increase**

DASHBOARD



Consists of three components

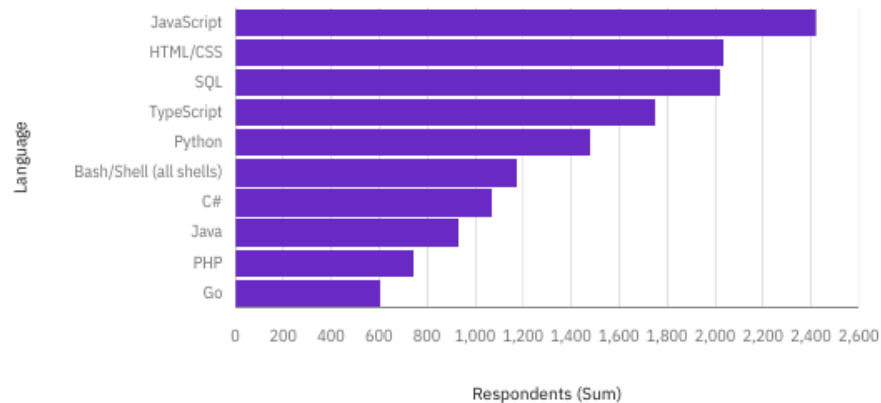
1. Current Technology Usage
2. Future Technology Trend
3. Demographics



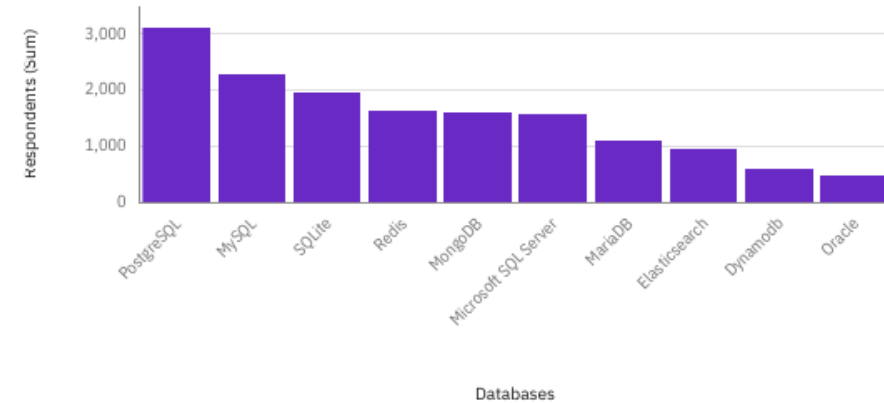
DASHBOARD TAB 1: Current Technology Usage

Current Technology Usage

Top 10 Programming Languages Used



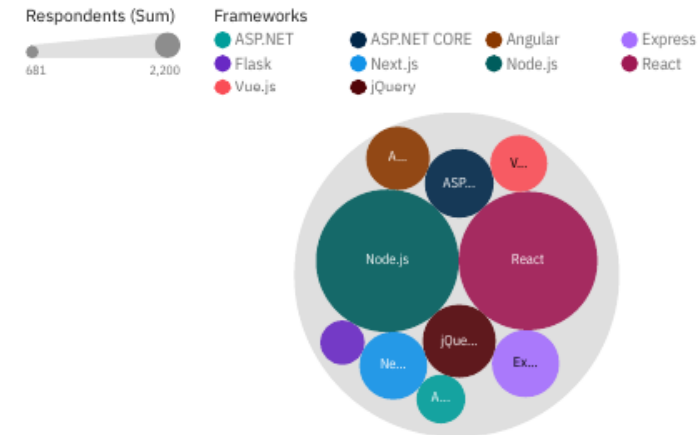
Top 10 Databases Used



Top 10 Platforms Used



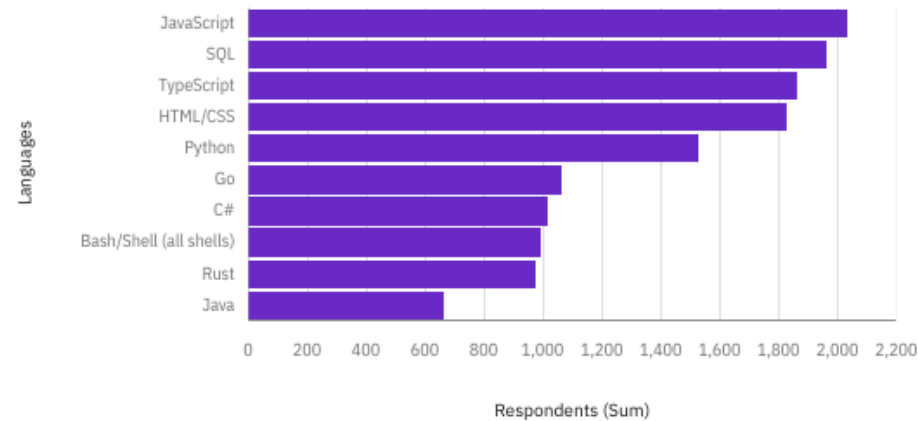
Top 10 Web Frameworks Used



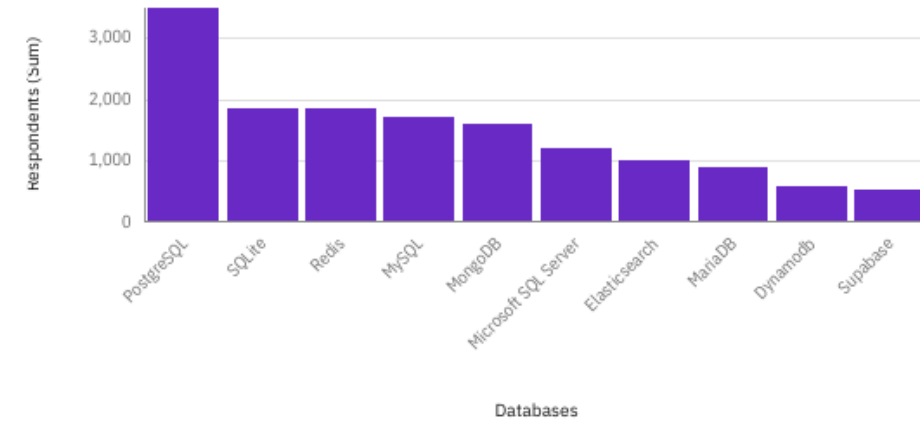
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trend

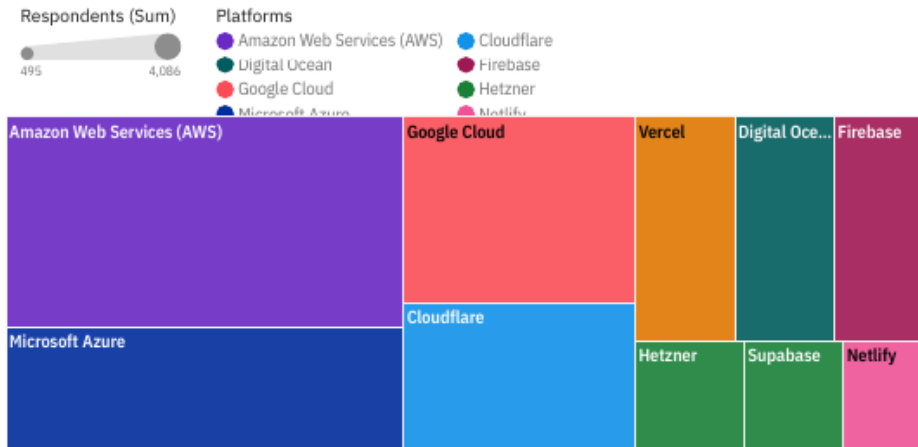
Top 10 Desired Programming Languages



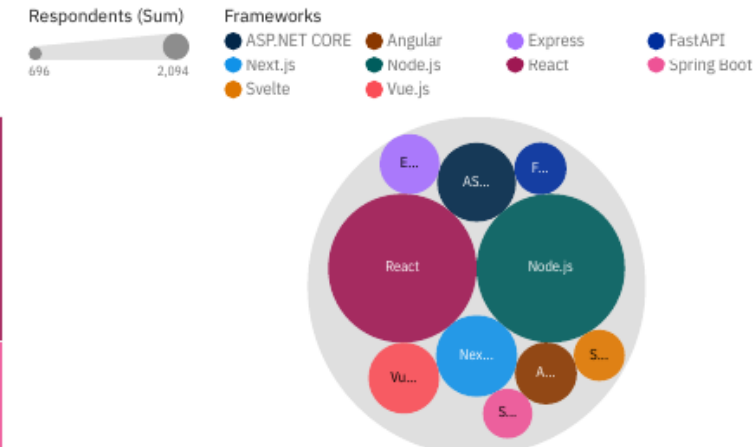
Top 10 Desired Databases



Top 10 Desired Platforms



Top 10 Desired Web Frameworks



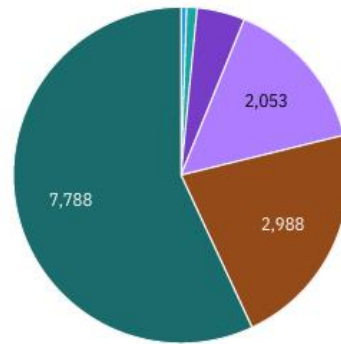
DASHBOARD TAB 3: Demographics

Demographics

Respondents by Age

Age

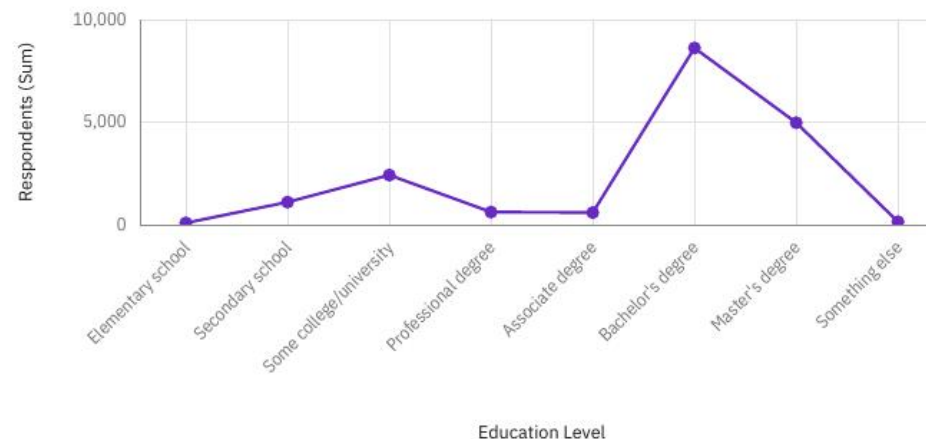
- 65 years or older
- Under 18 years old
- 55-64 years old
- 45-54 years old
- 18-24 years old
- 25-34 years old



Respondents by Country



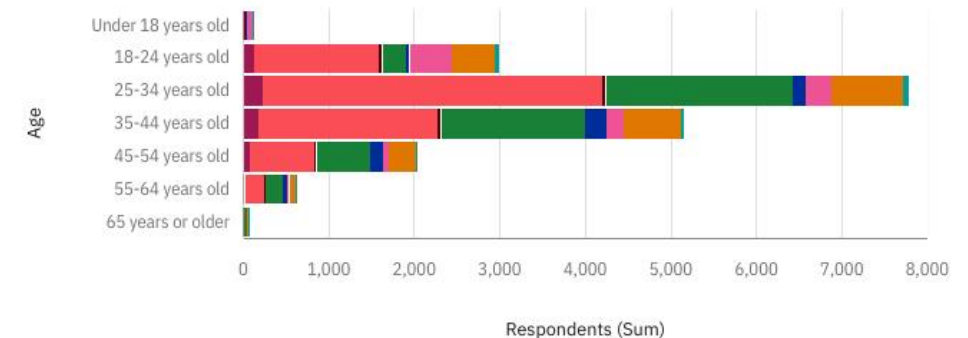
Respondents by Education Level



Respondents by Age and Education Level

Education Level

- Associate degree
- Bachelor's degree
- Elementary school
- Master's degree
- Professional degree
- Secondary school
- Some college/university
- Something else



DISCUSSION

- Discuss overall insights derived from dashboards



OVERALL FINDINGS & IMPLICATIONS

Findings

- Many of the top technologies are **Web Based**
- There **isn't much difference** between current use and desired technologies

Implications

- **Web and Cloud** based technologies remain the **most in-use**
- There is a **consensus** between what people are using and what they want to use

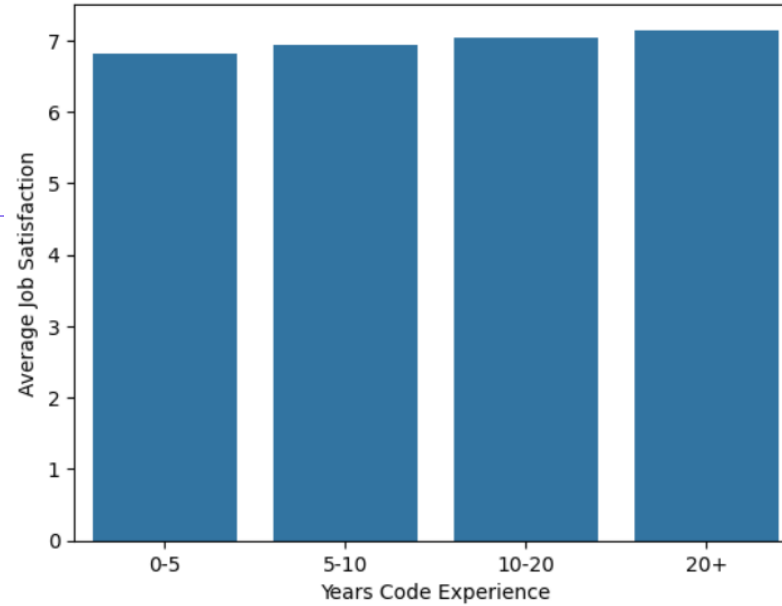
CONCLUSION



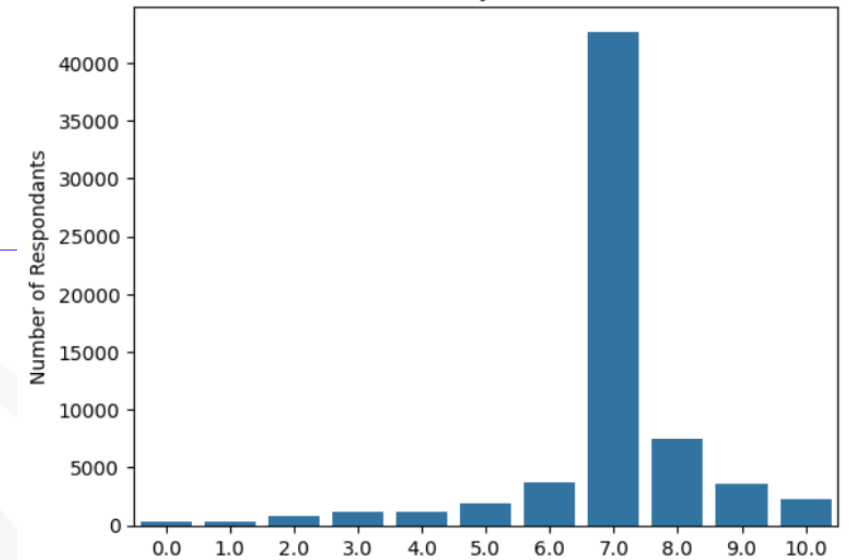
- **Web and Cloud technologies** reign supreme
- **Relational databases** are still the most widely used
- There is consensus between used and desired technologies

APPENDIX

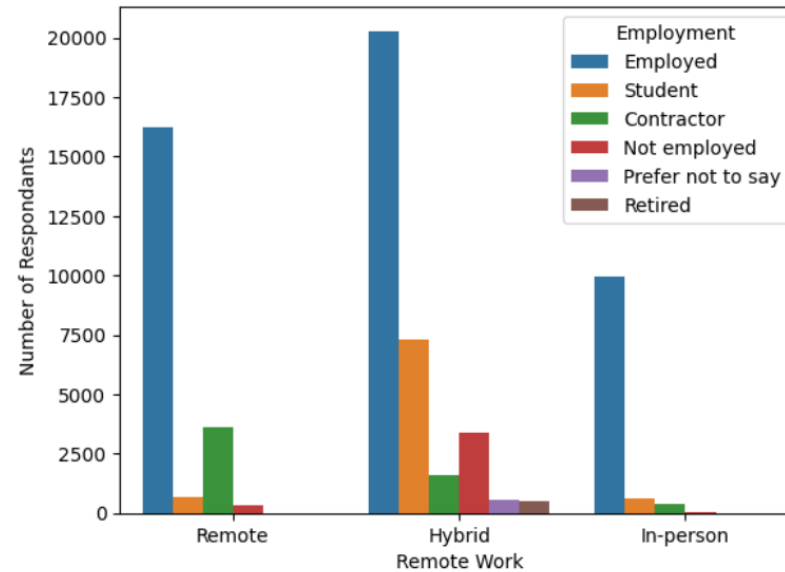
Average Job Satisfaction by Years of Coding Experience



Distribution of Job Satisfaction



Distribution of Remote Work



Distribution of respondents by salary

